Joseph Mangapit

(240)-274-1367 | joseph.n.mangapit@gmail.com | linkedin.com/in/joseph-mangapit/ | github.com/Badmininja

Education

University of California, Riverside — Riverside, CA
Master of Science in Computer Science — *3.56*

University of California, Riverside — Riverside, CA
Bachelor of Science in Computer Science with Business Applications — *Cum Laude 3.78*

Technical Skills

Languages: Python, C++, Java, JavaScript, SQL, HTML/CSS
Technologies: OpenAI API, REST APIs, Vicuna, PostgreSQL, Apache Spark
Libraries/Frameworks: PyTorch, scikit-learn, Pandas, Matplotlib, Vue.js, FastChat, WandB, Hugging Face
Developer Tools: Docker, Git/GitHub, Ghidra, Jupyter, Google Colab, VS Code, Visual Studio
Software Practices: Machine Learning, NLP, Unit Testing (GoogleTest), Database Design, SDLC, Agile/Scrum

Experience

Software Engineer Intern — Sep. 2023 – Dec. 2024
BitSecurerLab — *Riverside, CA*

- Built an interactive **3D force-directed call graph** in **Vue.js** (**JavaScript**/**HTML**/**CSS**) with 3d-force-graph and Three.js to visualize function and memory relationships from **Ghidra** data served over a **REST API**, improving analysis clarity by 25%.
- Implemented graph interactions including neighbor highlighting with animated edges, multi-node selection, function search with **autocomplete** and camera fly-to, and a popup subgraph view for memory objects.
- Developed features on **Git** branches while pair-programming in an **Agile** team, contributing to an online disassembler built on **Ghidra** and applying reverse engineering to interpret binary structures.

Teacher Assistant (AI) — Mar. 2024 – Jun. 2024
University of California, Riverside — *Riverside, CA*

- Increased student performance by up to 30% by leading weekly discussion sessions for 30+ students, explaining core AI concepts such as **supervised learning and data pre-processing**.
- Helped develop course materials, coding examples, and labs in **Python** and **Jupyter Notebooks**.
- Held weekly office hours with 1-on-1 help on **Python** debugging, **data pipelines**, and **machine learning** concepts.

Cage Operations Technician — Sep. 2026 – Present
Yaamava Resort & Casino — *Highland, CA*

- Maintains and repairs kiosks, ATMs, and currency counters while following gaming compliance procedures.

Projects

Automatically Jailbreaking Black-Box LLMs | *Python, OpenAI API, FastChat, WandB, Vicuna*

- Built an **end-to-end Python** pipeline around the **Tree of Attacks with Pruning (TAP)** framework to generate and score jailbreak prompts against **Vicuna** using **OpenAI API** attacker/evaluator models, reaching a 44% success rate.
- Ran **AdvBench** experiments tuning branching factor, depth, and width, and tracked judge scores and strategy comparisons across **LLM APIs** with **Weights & Biases**.

BrewDB — Café Management System | *Java, JDBC, PostgreSQL, Docker*

- Designed a normalized four-table **PostgreSQL** schema with **indexing** and built a role-based **Java**/**JDBC** command-line application (customer and manager modes) over the live database.
- Loaded a seed dataset of 28k users / 87k orders / 130k records and **containerized** the app and database with **Docker Compose**, replacing a hardcoded lab setup so it runs anywhere.

Yelp's Perfect Establishments | *Python, Apache Spark, scikit-learn, Docker*

- Wrote **Spark SQL** queries joining review and business data and built per-state **Matplotlib**/**Seaborn** visualizations of star ratings and review attributes with **PySpark** and **Pandas**.
- Collaborated in a four-person team on **Spark** analysis, a **Random Forest** model, and an interactive map; **containerized** the pipeline with **Docker** for reproducible one-command runs.